

Adaptive Night Light

Measure, decide, illuminate, and explain why

The circuit makes its internal decision visible without requiring Serial.

Lesson	009 — integration project	Time	150–210 minutes
Prerequisites	Lessons 001–008	Board	Arduino Mega 2560
Evidence	voltage, raw sample, mode, PWM	Status	host verified; bench open
Safety	E1: USB-powered passive sensor and current-limited LEDs only		

1 Mission and evidence chain

Build a night light whose behavior can be explained at every layer:

ambient light → A0 voltage → raw sample → filtered permille → hysteresis mode → D9 PWM.

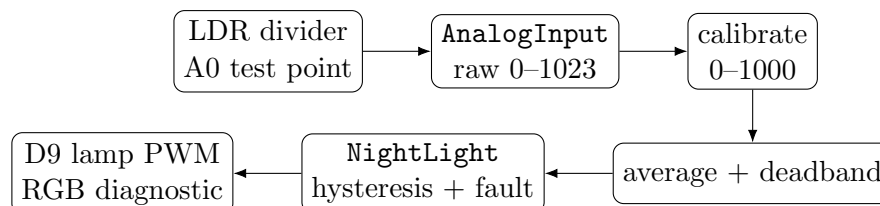
The white lamp is the primary output. A common-cathode RGB LED is the diagnostic output: blue means **Ready**, green means **Active**, and red means **SensorFault**. The named sensor test point is the divider junction at A0; the named safe-state test point is D9 relative to GND.

By the end, you can:

- compose `AnalogInput`, calibration, moving average, deadband, `NightLight`, `PwmOutput`, and `RgbLed`;
- explain why calibration, filtering, deadband, and hysteresis solve different problems;
- reproduce a decision from an ordered sample trace;
- distinguish sensor open/short evidence from ordinary darkness or light;
- prove resource acquisition separately from the lamp's safe state;
- build, upload, monitor, and capture optional logs entirely from the CLI.

Safety checkpoint. Disconnect USB and all other power before wiring. Use one 330Ω resistor for the white LED and one for each RGB die. The sensor divider must remain between the Mega's 5 V and GND rails; never apply an external signal to A0. Stop for heat, odor, resets, flicker unrelated to the lesson, or any measured voltage outside 0–5 V.

2 System composition



The canonical sketch deliberately reads like this diagram. Its `loop()` first calls `observeLight()`, then `decideLighting()`, then `actuateLighting()`. Hardware ownership remains in the endpoints; `NightLight` receives a value and knows nothing about Arduino pins.

2.1 Why four transformations?

Stage	Question answered	What it does not prove
Calibration	What normalized light does this raw value represent?	Stability over time
Moving average	What recent level does a noisy stream support?	A mode decision
Deadband	Is this tiny change worth reporting?	Lamp on/off policy
Hysteresis	Has light crossed the appropriate boundary for the current mode?	Electrical output

3 Orientation drawing

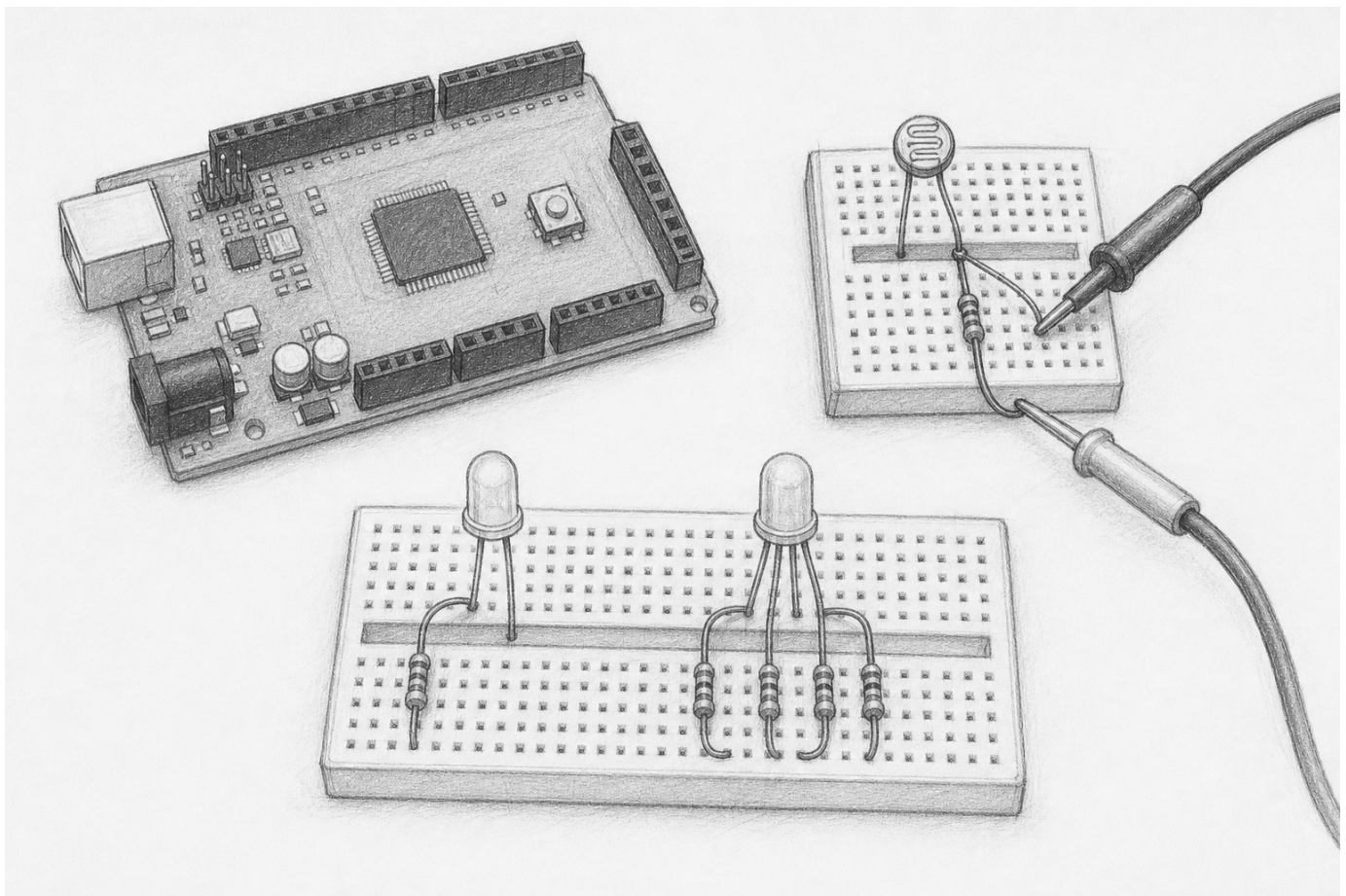


Figure 1: Pencil orientation showing the sensing area, Mega, indicator area, and meter probes. It is not an authoritative wiring diagram; positions, breadboard connectivity, and lead order are simplified. Build only from the literal schematic and connection table below.

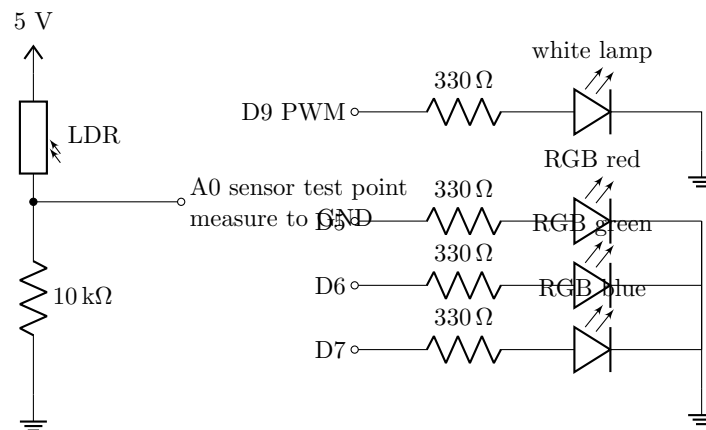
4 Parts and literal wiring

Quantity	Part	Requirement
1	Mega 2560	USB powered
1	photoresistor	identified passive LDR; record its datasheet or kit reference
1	resistor	10 k Ω , divider lower leg
1	white LED	identified polarity and forward-voltage range
1	common-cathode RGB LED	verify package pinout; lead order is not universal
4	resistors	330 Ω , one per LED die
1	multimeter	DC voltage and unpowered resistance/continuity

Connections.

Function	Literal path	Observation
sensor	5 V $\rightarrow$ LDR $\rightarrow$ A0 junction $\rightarrow$ 10 k Ω $\rightarrow$ GND	A0 to GND: 0–5 V
lamp	D9 $\rightarrow$ 330 Ω $\rightarrow$ white LED anode; cathode $\rightarrow$ GND	D9 PWM test point
RGB red	D5 $\rightarrow$ 330 Ω $\rightarrow$ red anode	red fault pattern
RGB green	D6 $\rightarrow$ 330 Ω $\rightarrow$ green anode	green active pattern
RGB blue	D7 $\rightarrow$ 330 Ω $\rightarrow$ blue anode	blue ready pattern
RGB return	common cathode $\rightarrow$ GND	unpowered continuity to GND

The fixed 10 k Ω lower leg limits divider current to at most 0.5 mA at 5 V even if the LDR is shorted. Its Thevenin source impedance is never above 10 k Ω , matching the conservative ADC source-impedance guidance used in this lesson.



More light lowers LDR resistance, raises A0 voltage, and raises the normalized `lightPer mille` value. If your identified sensor behaves differently, change the divider orientation and calibration together; do not silently invert only the software.

4.1 Current and timer budget

For each LED die, calculate $I = (V_{OH} - V_f)/330\Omega$ with the actual part's minimum V_f and a conservative loaded V_{OH} . Record the worst simultaneous case: the lamp plus up to three RGB dies. Keep margin below every ATmega2560 pin, port-group, and package operating limit; absolute maximum ratings are not design targets. This lesson additionally requires the calculated and measured current to remain at or below 15 mA per channel.

D9 uses the Mega's Timer2 PWM resource. D5, D6, and D7 use other PWM timers. Do not add `PiezoSunder`

without resolving its exclusive Timer2 claim: the conflict must fail during initialization, before an active waveform.

5 Decision contract

```

adk::NightLightConfig nightLightConfig;
nightLightConfig.turnOnBelowPermille = 350;
nightLightConfig.turnOffAbovePermille = 450;
nightLightConfig.minimumDuty         = 24;
nightLightConfig.maximumDuty         = 192;

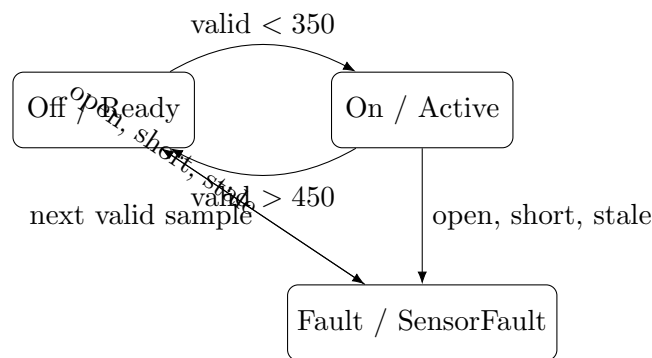
adk::NightLight nightLight (nightLightConfig);

const adk::NightLightInput observation (
    filteredLight,
    adk::LightSampleState::Valid);

nightLight.update (observation);
const adk::NightLightSnapshot decision = nightLight.snapshot ();

```

NightLight starts **Off**. A valid value strictly below 350 permille changes it to **On**; it remains on through the middle band and changes off only strictly above 450. Equality does not cross either boundary. This memory is hysteresis, not filtering.



Any explicit **BelowRange**, **AboveRange**, or **Stale** sample enters **Fault**, forces duty zero, and selects the red diagnostic. The next valid sample first recovers to **Off**; darkness must then satisfy the ordinary turn-on rule. This prevents a recovered sensor from inheriting an old active command.

6 Deterministic trace exercise

Use an eight-sample moving average and an eight-permille deadband. For this short worksheet, the “filtered” column is supplied so the state-machine reasoning remains visible.

Step	raw	filtered	state	prior mode	next mode	duty
1	700	680	Valid	Off	_____	_____
2	360	440	Valid	Off	_____	_____
3	240	340	Valid	Off	_____	_____
4	410	400	Valid	On	_____	_____
5	520	451	Valid	On	_____	_____
6	1023	—	AboveRange	Off	_____	_____
7	220	330	Valid	Fault	_____	_____

Replay the same ordered samples twice in the host test. Equal configuration and samples must produce equal snapshots. No test reads wall time, sleeps, or depends on Serial delivery.

7 Fault matrix

Fault or boundary	Expected decision	Lamp	RGB
A0 near 0	suspected divider fault; not ordinary darkness	duty 0	red
A0 near 1023	suspected divider fault; not ordinary daylight	duty 0	red
missing fresh value	Stale	duty 0	red
350 exactly while off	strict boundary does not turn on	duty 0	blue
450 exactly while on	strict boundary does not turn off	active	green
D9 already claimed	initialization fails before operation	high impedance	unavailable or inactive
Timer2 exclusively claimed	initialization fails before PWM	high impedance	inactive
reset or shutdown	endpoint releases ownership	high impedance	high impedance

Do not overclaim the RGB evidence. Blue proves that all objects initialized and the controller selected **Ready**. It does not prove D9 is electrically safe. Measure D9 to GND separately: zero duty while initialized should be near 0 V; after shutdown the endpoint contract is high impedance, which needs a resistance or externally biased voltage test with the board in the documented test condition.

8 CLI-only workflow

No IDE is required. From the repository root:

```
make bootstrap
make host-test
make arduino-Lesson009AdaptiveNightLight
make upload EXAMPLE=Lesson009AdaptiveNightLight PORT=/dev/ttyACM0
make monitor-describe PORT=/dev/ttyACM0
make monitor PORT=/dev/ttyACM0 BAUD=115200
make serial-log PORT=/dev/ttyACM0 BAUD=115200 \
    SERIAL_LOG=build/serial/lesson009.log
make lessons
make site
```

The circuit supplies the acceptance evidence even if **monitor** is never run. Serial is useful only for optional raw and filtered numeric context. The canonical example contains no required Serial protocol.

9 Bench procedure: predict, observe, interpret

9.1 Stage A — inspect without power

1. ☐ Confirm the exact Mega pins, every 330 Ω resistor, RGB common cathode, and LDR divider orientation.
2. ☐ Measure the divider's lower resistor: expected _____, observed _____.
3. ☐ Confirm no continuity from 5 V directly to GND and no LED die lacks its series resistor.
4. ☐ Predict the A0 voltage in bright and covered conditions.

9.2 Stage B — acquisition evidence

Power from USB. Do not upload until the unpowered checks pass.

1. ☐ Build and upload with the CLI targets above.
2. ☐ Predict: valid moderate light selects blue and lamp off.
3. ☐ Observe RGB: _____.
4. ☐ Interpret: _____.

9.3 Stage C — sensor and decision evidence

1. ☐ Measure A0 to GND under bright light: _____ V. Cover the LDR and measure: _____ V.
2. ☐ Move light slowly through the on boundary. Record A0 voltage when the lamp becomes active: _____ V.
3. ☐ Reverse direction. Record voltage when it becomes ready: _____ V. Explain the different boundaries: _____.
4. ☐ Observe green with the lamp waveform. At D9, record PWM high level _____ V and approximate duty _____%.

9.4 Stage D — fault and safe-state evidence

Remove USB before changing the sensor connection. Simulate only one fault at a time, then restore wiring unpowered.

1. ☐ Open the LDR's 5 V leg while leaving A0 joined to the fixed lower resistor. Predict red and lamp duty zero. Observe: _____.
2. ☐ Restore, then short only the LDR terminals so A0 approaches 5 V through the identified topology. Predict red and duty zero. Observe: _____.
3. ☐ Restore normal wiring. Confirm recovery goes through blue before darkness can select green.
4. ☐ Record D9 initialized at zero duty: _____ V. Run a controlled shutdown build or acceptance fixture, then measure the documented high-impedance condition: _____.

This procedure detects the named open and short simulations because the fixed divider leg drives A0 toward a rail. A broken A0 jumper can float at a plausible value and is not universally detectable with one ADC channel. Even then, every accepted sample remains bounded and output duty cannot exceed the configured maximum; inspect continuity unpowered when evidence disagrees.

10 Diagnosis by layer

Observation	Most useful next check	Reason
no RGB at startup	resource conflict, polarity, resistor, common cathode	acquisition evidence is absent
red RGB, lamp off	A0 voltage and divider continuity	controller reports sensor evidence invalid
blue/green changes, lamp dark	D9 waveform, LED polarity, lamp resistor	decision exists; actuator path is suspect
lamp chatters near threshold	raw range, average window, dead-band, threshold order	separate noise from policy
A0 changes, mode does not	calibration direction and normalized values	electrical sensing works; mapping may not
unexpected reset	disconnect power and inspect shorts/current budget	software evidence cannot clear an electrical fault

11 Design exercises

1. Plot measured A0 voltage against normalized permille at five light levels. Which calibration points should be learned from your room?
2. Change only the moving-average window. Predict latency and noise before compiling; then record both.
3. Change only the hysteresis gap. Find the smallest gap that avoids chatter in a repeatable hand-shadow trace.
4. Propose a stale-sample policy using explicit time. State the maximum valid age and add rollover cases before changing the interface.
5. Add a host trace for one sensor fault during active PWM. Prove the first fault snapshot commands zero duty and the first recovery snapshot is `Off`.

12 Acceptance record

Field	Record	Field	Record
Board/revision		Date	
USB supply		Reviewer	
Arduino CLI/core		ADK revision	
Multimeter		LDR identity	
White LED V_f		RGB identity	
Flash bytes		Static RAM bytes	

- ☐ Unpowered wiring and current calculations passed.
- ☐ Ready, Active, and SensorFault indications matched the table.
- ☐ A0 voltage supported the raw-sample interpretation.
- ☐ Hysteresis produced distinct on and off boundaries.
- ☐ Open and short sensor tests forced lamp duty zero.
- ☐ Resource acquisition and D9 safe state were proven separately.
- ☐ Reset, shutdown, and power removal observations were recorded.

Observed deviations and unresolved checks:

Release status. Compilation, host replay, and a generated PDF do not constitute physical verification. Until a person completes this card on the named Mega 2560 and records measurements, this lesson remains host verified with hardware acceptance open.

13 Use with

Use the maintained HTML lesson for searchable API links, errata, source downloads, and exact CLI targets. Use this PDF at the bench for predictions, schematics, worksheets, and recorded evidence. The canonical editable sketch is `examples/Lesson009AdaptiveNightLight/` rather than a duplicate in this document.

- Lesson 009 HTML
- Canonical sketch
- NightLight API
- Arduino Mega 2560 datasheet
- ATmega2560 primary product page